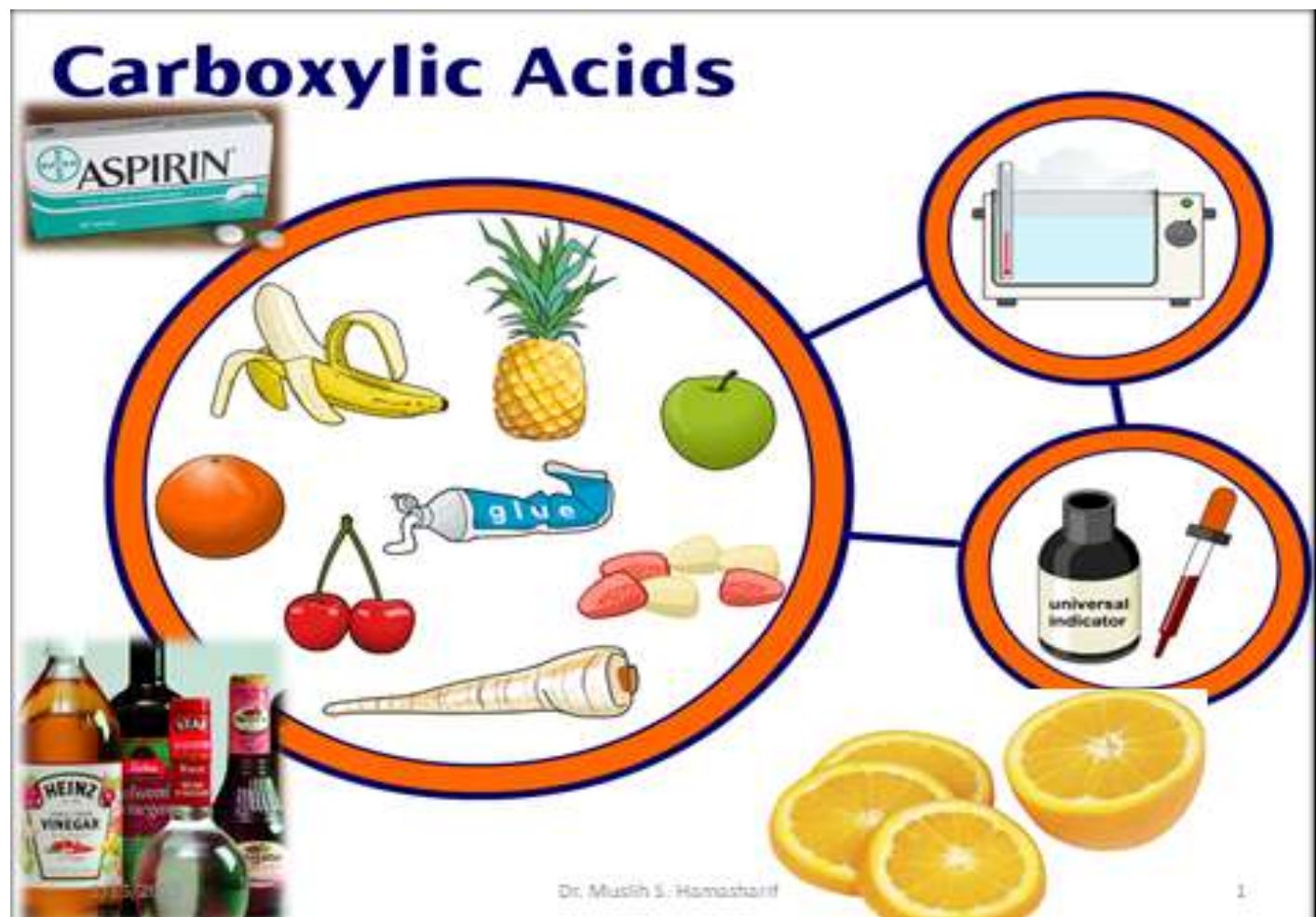




Some Important Acids

- Acetic acid is in vinegar and other foods, used industrially as a solvent, catalyst, and reagent for synthesis.
- Fatty acids from fats and oils.
- Benzoic acid is found in drugs and preservatives.
- Adipic acid is used to make nylon 66.
- Phthalic acid is used to make polyesters.

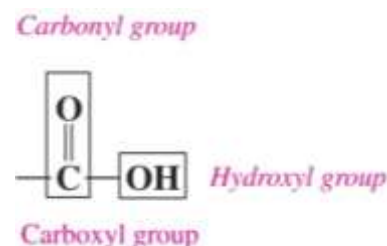
Carboxylic acids in soaps

The base hydrolysis of long chain fatty acids produces acid salts called “soaps”.

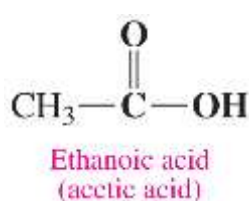


1-Introduction:

- A carboxylic acid contains a hydroxyl group (–OH) attached to a carbonyl group.

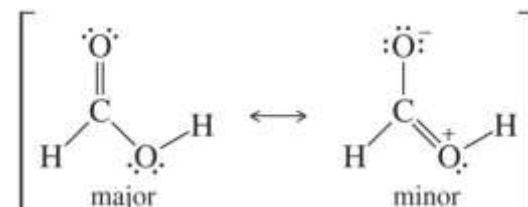


- Carboxylic acids are classified as aliphatic or aromatic depending on whether (R) or an (Ar) is attached to the carboxylic group.



- Resonance Structures of Formic Acid

One of the unshared electron pairs on the hydroxyl oxygen atom is delocalized into the electrophilic pi system of the carbonyl group, O—H eclipsed with C=O, to get overlap of π orbital with orbital of lone pair on oxygen.

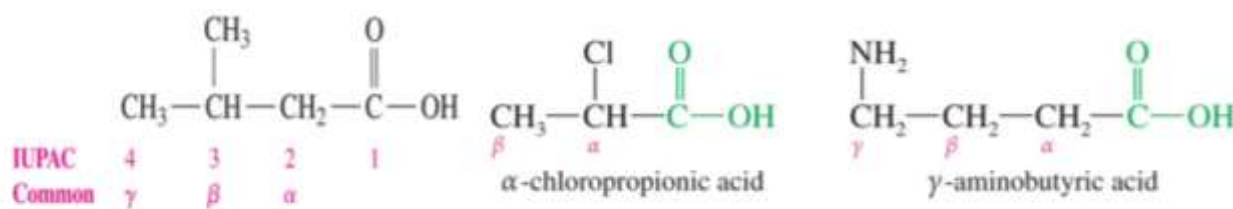


2. Naming the carboxylic Acids

2.1. Common Names:

The common names of simple carboxylic acids

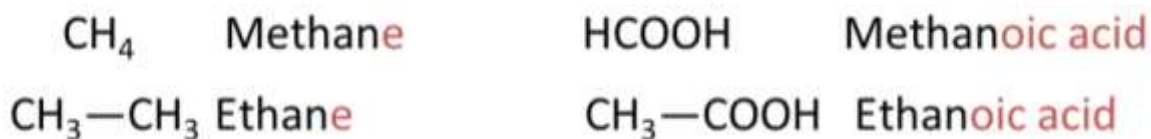
- ❖ Are formic acid (1C), acetic acid (2C), propionic acid (3C), and butyric acid (4C).
HCOOH formic acid, CH₃COOH acetic acid
- ❖ Locate substituents using α , β , γ for the carbon atoms adjacent to the carboxyl carbon.



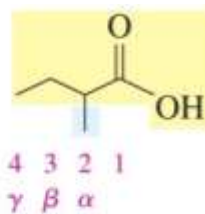
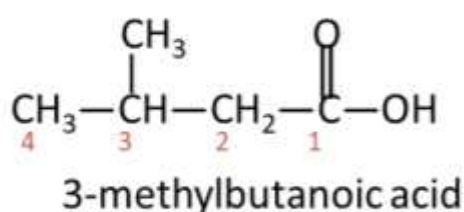
~~3-methylbutanoic acid~~
(β -methylbutyric acid)

2.2. The IUPAC Names for Carboxylic Acids:

1. Identify the longest carbon chain and replace the **-e** in the alkane name with **-oic acid**.



2. Give the location and name of each substituent by counting the **carboxyl carbon as 1**



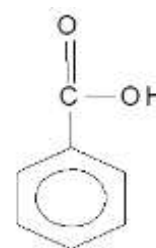
2-methylbutanoic acid
 α -methylbutyric acid

2.3. Naming Aromatic Carboxylic Acids:

Benzoic acid is the aromatic carboxylic acid.

Locates substituents by assigning 1 to the carbon attached to the carboxyl group.

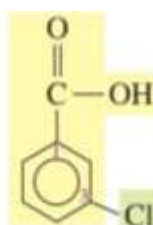
- Has common names that assign prefixes *ortho*, *meta*, and *para* for 2 substituents.



ortho 1, 2 location

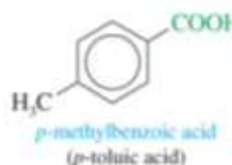
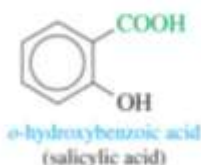
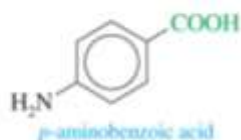
meta 1, 3 location

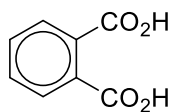
para 1, 4 location



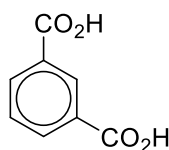
3-chlorobenzoic acid

meta-chlorobenzoic acid

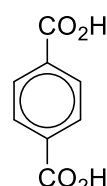


2.4-Dicarboxylic acids:

phthalic acid
Benzene-1,2-dicarboxylic acid



isophthalic acid
Benzene-1,3-dicarboxylic acid



terephthalic acid
Benzene-1,4-dicarboxylic acid

	<u>IUPAC Names</u>	<u>Common Names</u>
HOOC-COOH	Methanedioic acid	Oxalic acid
HO ₂ C-CH ₂ -CO ₂ H	Propanedioic acid	Malonic acid
HO ₂ C-CH ₂ CH ₂ -CO ₂ H	Butanedioic acid	Succinic acid

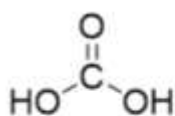
2.5-Salts of Carboxylic acids

Name of cation + Name of acid: drop -ic acid, add -ate

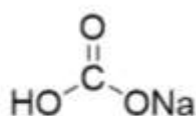
CH₃CO₂Na sodium acetate or sodium ethanoate

CH₃CH₂CH₂CO₂NH₄ ammonium butyrate
ammonium butanoate

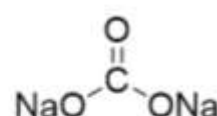
(CH₃CH₂COO)₂Mg magnesium propionate
magnesium propanoate



carbonic acid



sodium bicarbonate
sodium hydrogen carbonate
NaHCO₃

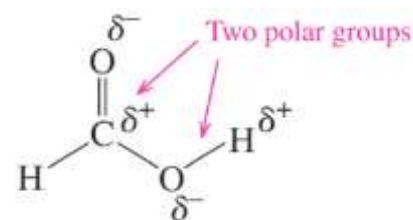


sodium carbonate
Na₂CO₃

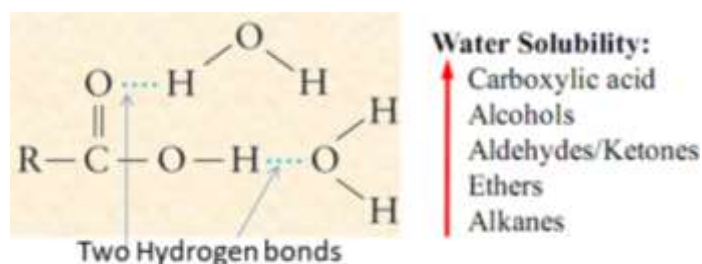
3. Physical Properties of carboxylic acids

3.1. Solubility of Carboxylic Acids:

- Are Carboxylic Acids strongly polar.(why?)
- Have two polar groups:
1-hydroxyl (–OH).
2-carbonyl (C=O).



The smaller carboxylic acids (1-4 carbons) are quite soluble in water.

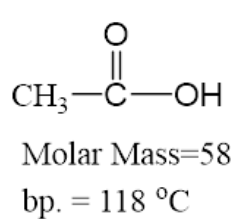
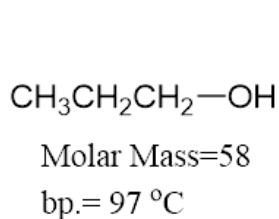
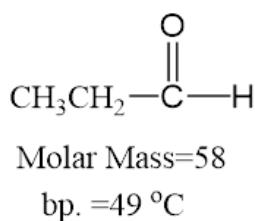
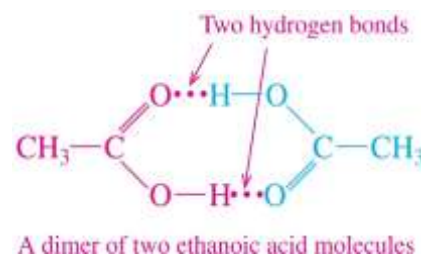


- As the number of carbons in a carboxylic acid series becomes greater, the solubility in water decreases.
- Aromatic carboxylic acids are insoluble in water.

3.2. Boiling Points of Carboxylic Acids:

The **boiling points of carboxylic acids** Are higher than alcohols, ketones, and aldehydes of similar mass. Why?

A//Carboxylic acids form **very strong** hydrogen bonds with themselves in the **absence of water** because they form **dimers** in which hydrogen bonds form between the polar groups in the two carboxyl groups.



Boiling Point:

- Carboxylic acid
- Alcohols
- Aldehydes/Ketones
- Ethers
- Alkanes

4. Preparation of Carboxylic acids:

4.1. Oxidation:

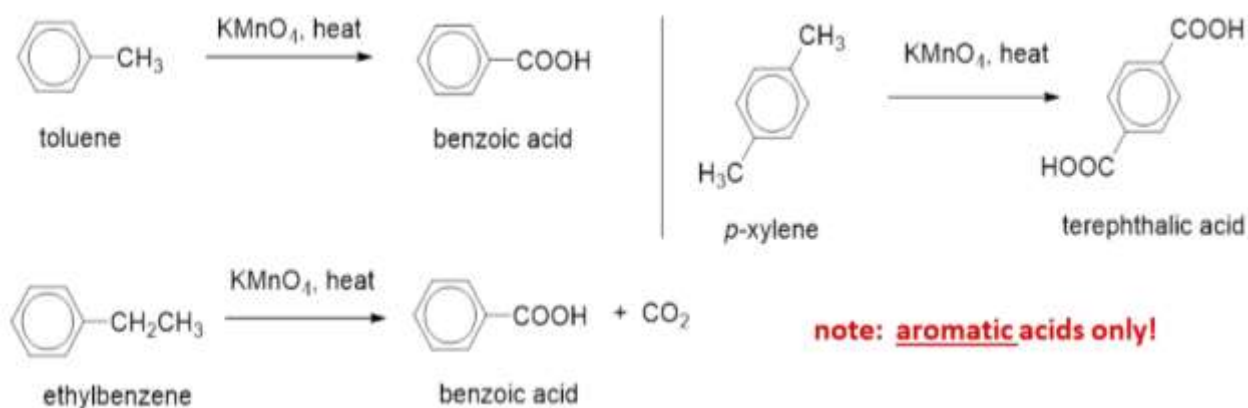
A. Oxidation of primary alcohols and aldehydes:

In chapter Alcohols, we learned- Oxidation of primary alcohols and aldehydes-for the preparation of Carboxylic acids from alcohols and Aldehydes



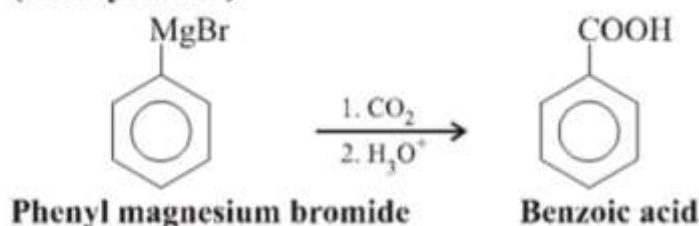
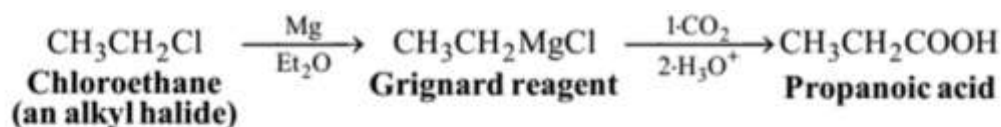
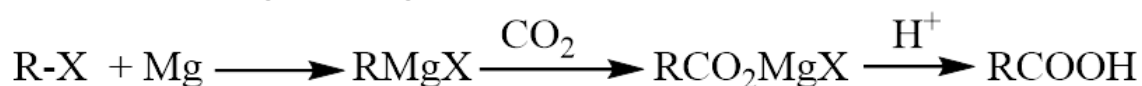
B. oxidation of arenes:

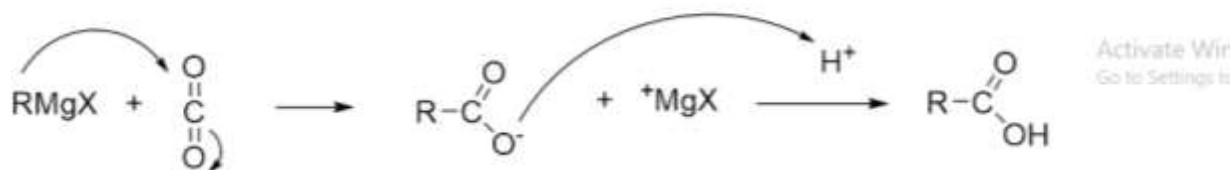
In chapter Arenes, we learned- Oxidation of Arenes- preparation of Carboxylic acids from Arenes.



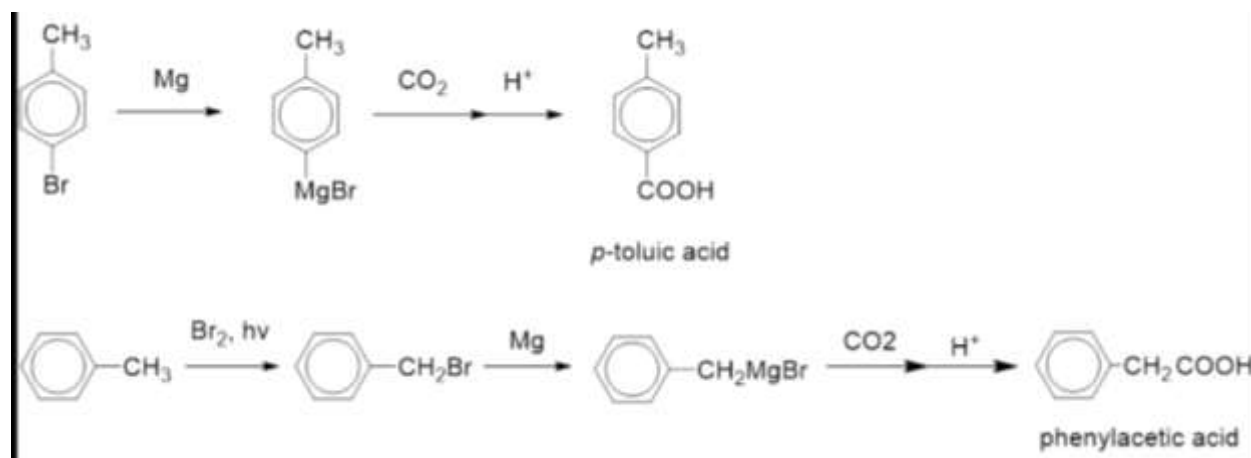
4. 2. Carbonation of Grignard reagent:

Grignard reagents react with carbon dioxide to form carboxylic acids which have one carbon more than than Grignard reagent.



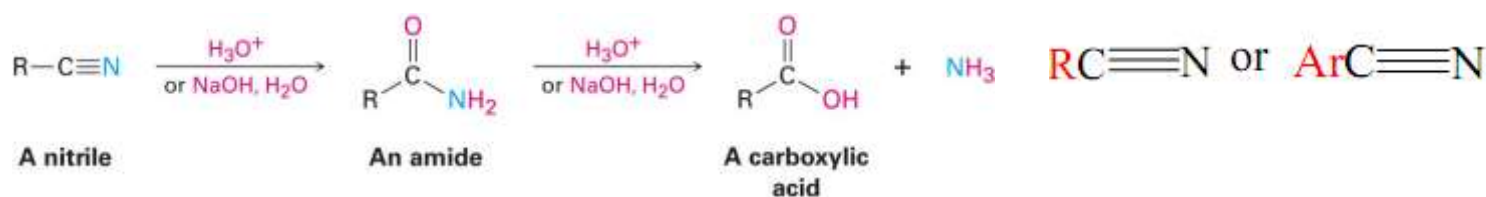
Mechanism:

Q// How to prepare From.....?

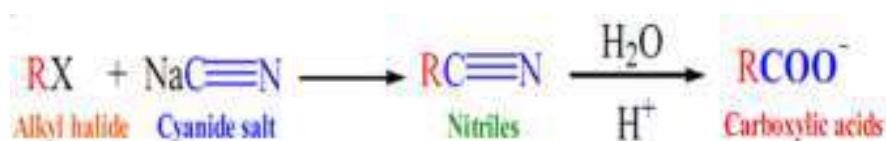
**4.3. Hydrolysis of a nitrile:**

When a nitrile (alkyl cyanide) that the nitrile (-CN) group is converted to the carboxyl (-COOH) group.

❖ Occurs in both acidic and basic environments.



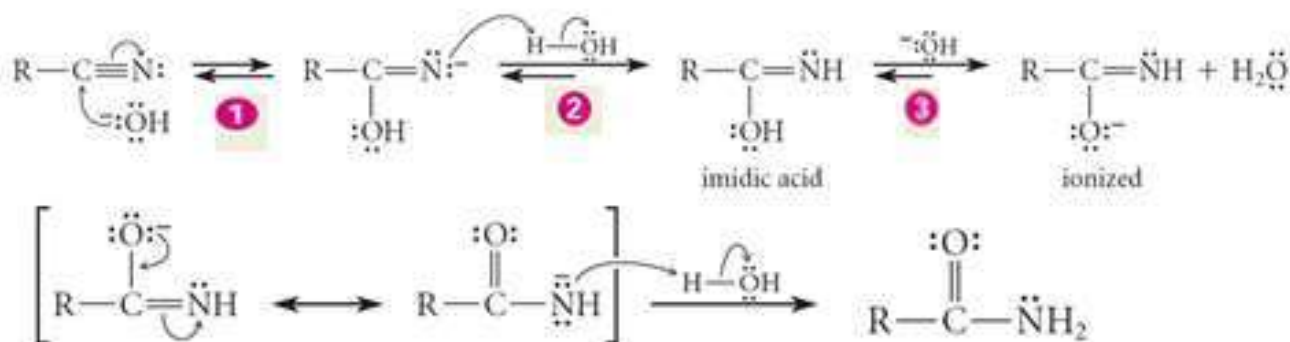
- They are prepared by reacting a 1° or 2° alkyl halide with cyanide salt.
- Acid hydrolysis of a nitriles yields a carboxylic acids.



Mechanism of Hydrolysis of Nitriles:

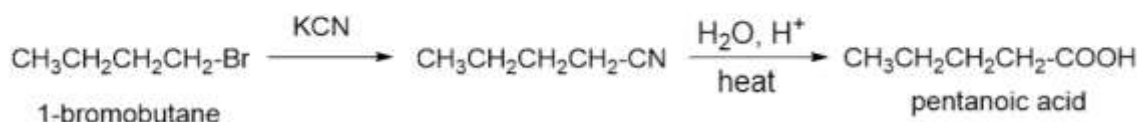
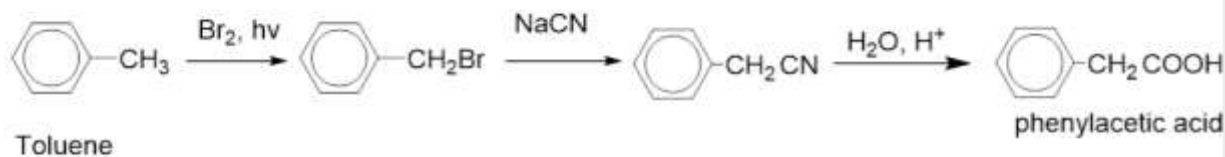
1-Nucleophilic addition of hydroxide to C≡N bond.

2 and 3- Protonation gives a hydroxy imine, which tautomerizes to an amide.



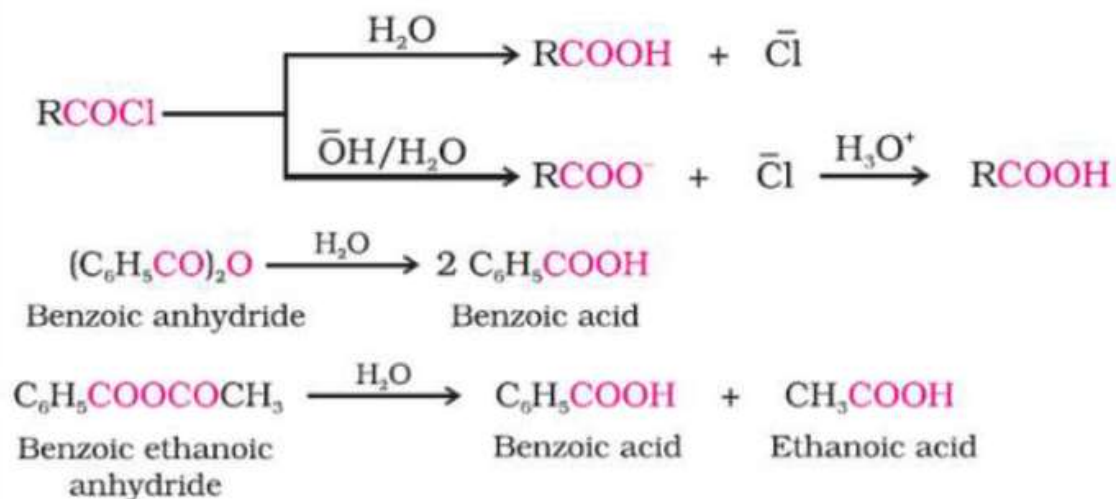
4- A/ A second hydroxide adds to the amide carbonyl group and loss of a proton gives a dianion.

B/Expulsion of NH_2^- gives the carboxylate.

**Acidic hydrolysis:****Basic hydrolysis:****➤ How to Prepare Carboxylic Acid from:**

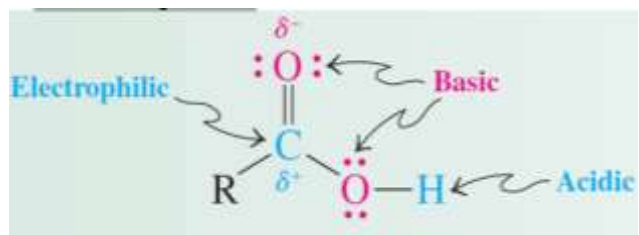
4.4. Preparation carboxylic acid by Acyl Halides and Anhydrides:

When acid chloride is hydrolysed with water then carboxylic acid is produced and it is more readily hydrolysed with an aqueous base and gives corresponding carboxylic ions which on further acidification gives the carboxylic acid.



5. Reaction of carboxylic acids:

Much of the functional behavior of carboxylic acids can be anticipated if we view them structurally as hydroxy carbonyl derivatives. Thus, the hydroxy hydrogen is acidic, the oxygens are basic and nucleophilic, and the carbonyl carbon is subject to nucleophilic attack.



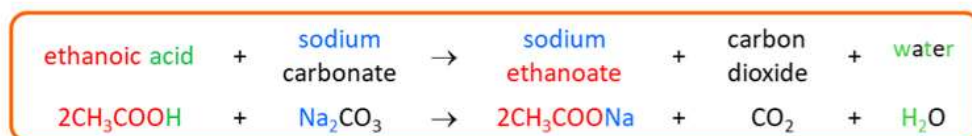
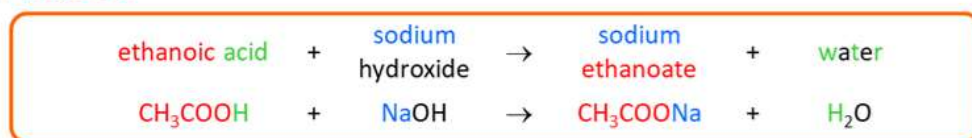
5.1. Acidity of Carboxylic Acids:

Carboxylic acids react like any other type of acid, forming salts when they react with alkalis, carbonates and metals: **Identification**)

a) with active metals:



b) with bases:



5.1. Acidity of Carboxylic Acids: (Homework)



$$K_a = \frac{[\text{R}-\text{CO}_2^-][\text{H}_3\text{O}^+]}{[\text{R}-\text{CO}_2\text{H}]}$$

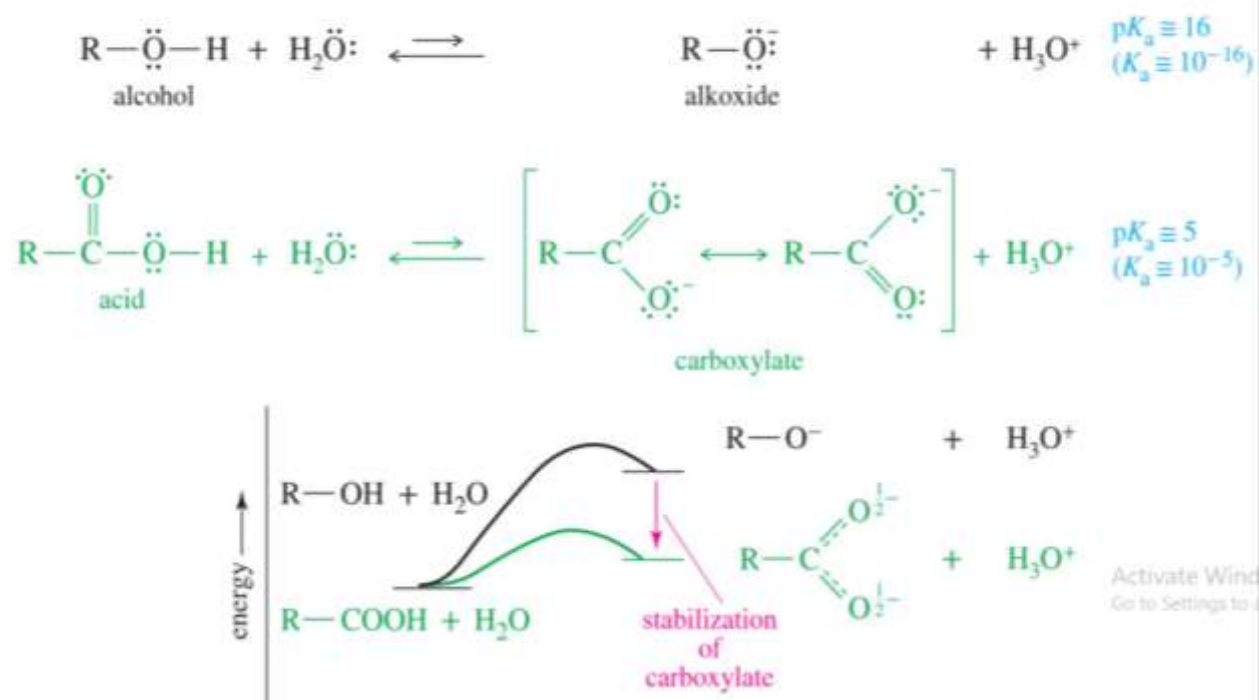
$$\text{p}K_a = -\log_{10} K_a$$

- Carboxylic acids are weak acids.
- A carboxylic acid may dissociate in water to give hydronium ions and a carboxylate ion.

- The equilibrium constant K_a for this reaction is called the **acid-dissociation constant**.

Q//Why are carboxylic acids more acidic than alcohols?

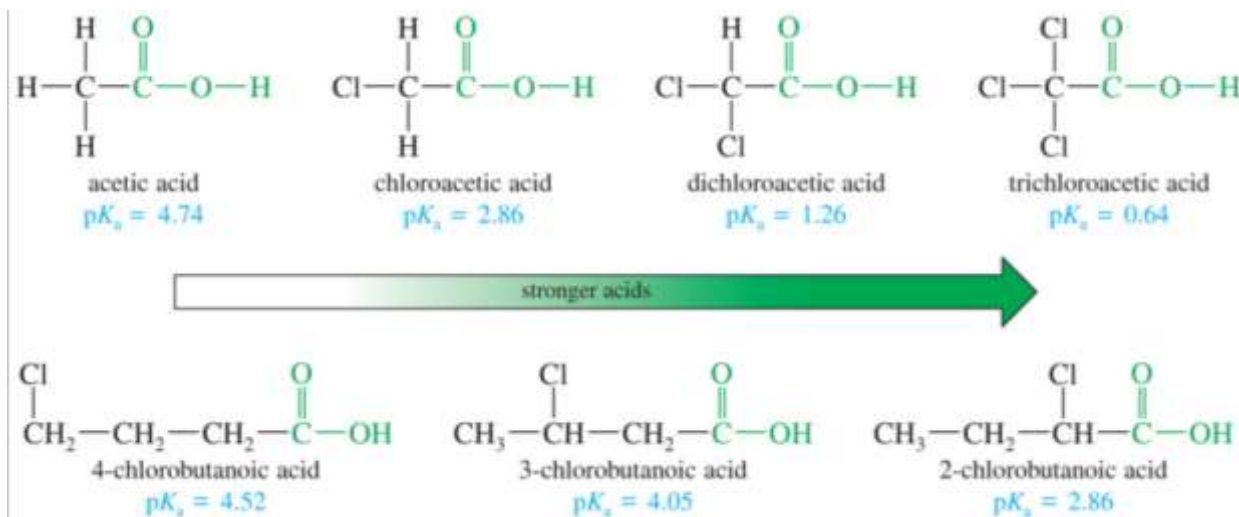
Energy Diagram of Carboxylic Acids and Alcohols



Effect of substituent groups on acid strength?

- Any factor that stabilizes the anion more than it stabilizes the acid should increase the acid.
- Electron-withdrawing groups should disperse the negative charge, stabilize the anion, and thus increase the acid strength.
- Electron-donating groups should intensify the negative charge, destabilize the anion, and thus decrease the acid strength.

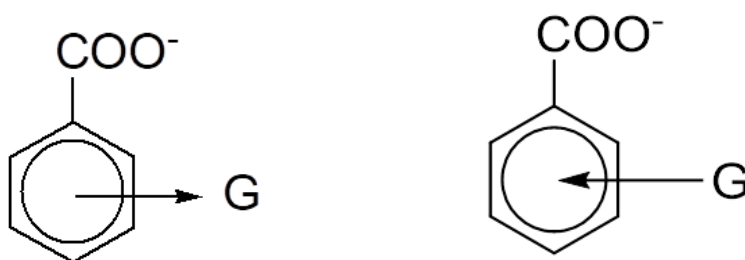
Resonance stabilization of the carboxylate ion, shifts the ionization in water to the right, increases the K_a , and results in carboxylic acids being stronger acids.



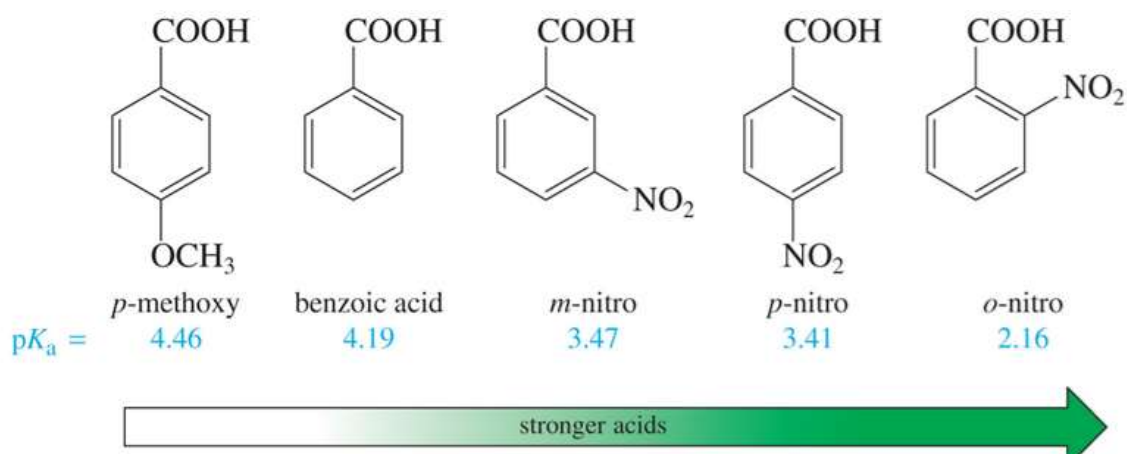
- Cl is electron withdrawing and delocalizes the negative charge on the carboxylate ion, shifting the ionization to the right and increase K_a also increasing acid strength.
- The magnitude of a substituent effect depends on its distance from the carboxyl group.
 - Inductive effect : Decrease with distance
 - Distance increase = decrease acidity

Effect of substituent groups on acid strength of benzoic acids?

- **Electron withdrawing groups** will stabilize the anion, shift the ionization to the right, increasing the K_a , increasing acid strength.
- **Electron donating groups** will destabilize the anion, increase the ΔH , shift the ionization in water to the left, decreasing the K_a , **decreasing acid strength**.



	<u>K_a</u>
p-aminobenzoic acid	1.4×10^{-5}
p-hydroxybenzoic acid	2.6×10^{-5}
p-methoxybenzoic acid	3.3×10^{-5}
p-toluic acid	4.2×10^{-5}
benzoic acid	6.3×10^{-5}
p-chlorobenzoic acid	10.3×10^{-5}
p-nitrobenzoic acid	36×10^{-5}

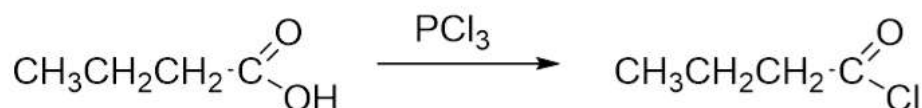
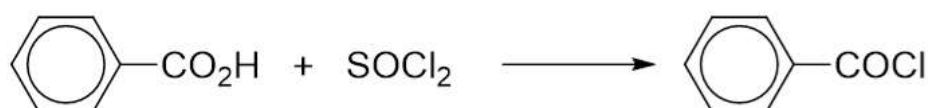
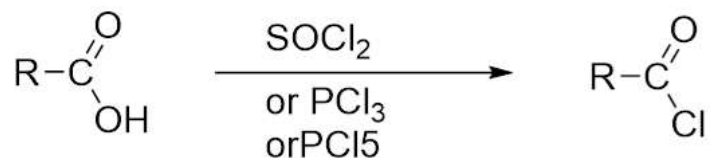


- Electron-withdrawing groups enhance the acid strength, and electron-donating groups decrease the acid strength.
- Effects are strongest for substituents in the *ortho* and *para* positions.

5.2. Conversion into functional derivatives:

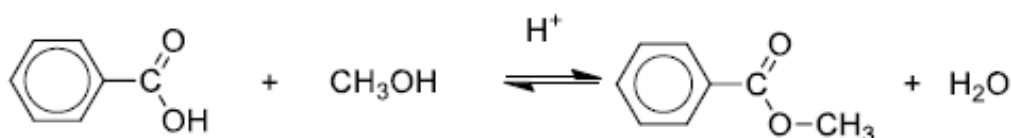
a) Conversion Carboxylic acid into acid chlorides.

Thionyl chloride, Phosphorus trichloride, or Phosphorus pentachloride.



b) carboxylic acid esters

A-direct" esterification:



- Reversible and often does not favor the ester.
- Use an excess of the alcohol or acid to shift equilibrium.
- Or remove the products to shift equilibrium to completion.

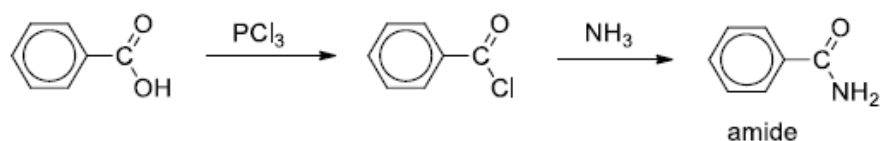
B-indirect" esterification:



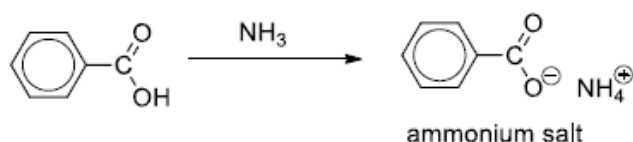
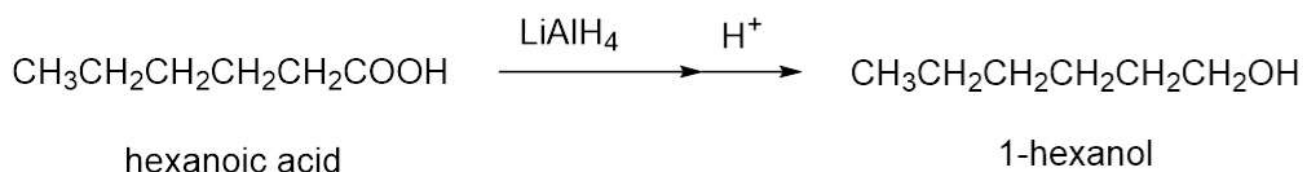
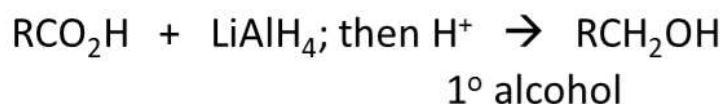
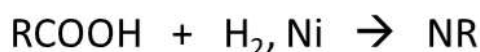
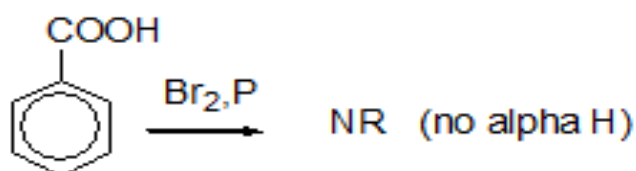
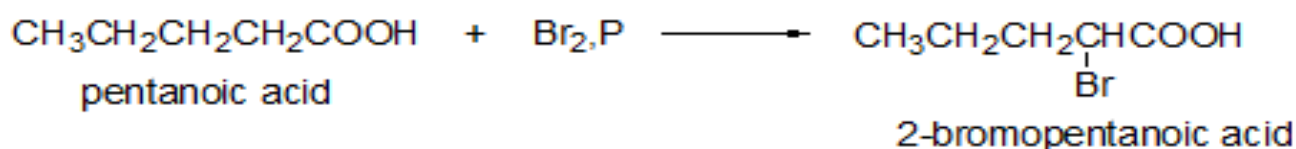
-convert the acid into the acid chloride first; not reversible

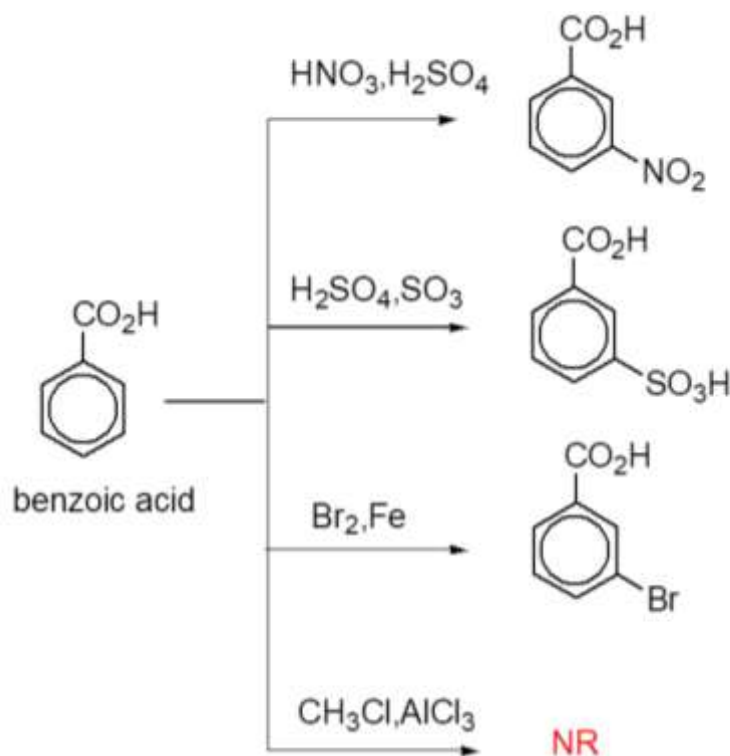
C. Carboxylic acid → amides

“indirect” only!



Directly reacting ammonia with a carboxylic acid results in an ammonium salt:

**5.3. Reduction:****Carboxylic acids resist catalytic reduction under normal conditions.****5.4. Alpha-halogenation: (Hell-Volhard-Zelinsky reaction)**

5.5. EAS: (-COOH is deactivating and *meta*- directing):

Activate Win
Go to Settings to

6. Identification- Carbonyl Compounds

Differentiation : to distinguish aldehydes from ketones, use a mild oxidising agent.

6.1. Litmus Paper -Acidity-

Turn blue litmus paper → red Litmus paper



Acid

Blue litmus turns red

6.2. Sodium Bicarbonate Test for Carboxylic Acids

Carboxylic acids can be identified with (sodium bicarbonate). Because carboxylic acids are stronger acids, they react with sodium bicarbonate. To produce carbon dioxide gas.

